



Department of Statistics and Data Science

Course Plan Odd Semester 2026-2027

MDS482-4 PROJECT

PROJECT PROPOSAL

PAWMAP

A Community-Driven Stray Animal Rescue and Welfare Platform

Submitted By

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1. Introduction

The issue of stray and abandoned animals in urban India has reached a scale that demands systematic, technology-driven intervention. With an estimated 62 million stray dogs and a large population of stray cats spread across cities and semi-urban localities, the welfare of these animals represents both a humanitarian concern and a public health challenge. Despite the presence of numerous non-governmental organisations (NGOs), animal welfare volunteers, and veterinary services, rescue and rehabilitation efforts remain fragmented, uncoordinated, and largely invisible to the broader community.

PawMap is a community-powered mobile and web application conceived to bridge this coordination gap. By placing real-time reporting, structured volunteer management, veterinary record-keeping, and adoption facilitation within a single, accessible interface, PawMap transforms passive bystanders into active participants in animal welfare. The platform is designed for scalability across Indian cities and draws on established technologies in cloud computing, geospatial services, and mobile application development to deliver a reliable and intuitive user experience.

The project is being developed as part of the MDS482-4 Project course under the Department of Statistics and Data Science, Christ University, Bengaluru. It represents the application of data science, software engineering, and community systems thinking to a domain of significant social relevance.

2. Need and Motivation

The motivation for PawMap arises from three converging realities observed in the urban Indian context:

2.1 Scale of the Problem

India hosts one of the largest stray animal populations in the world. The Animal Welfare Board of India and multiple municipal surveys indicate that stray dog populations in cities such as Bengaluru, Mumbai, and Delhi number in the hundreds of thousands. Abandoned pets, injured strays, and unvaccinated animals pose risks both to themselves and to the communities they inhabit. Current municipal responses are slow and under-resourced, while NGO efforts, though dedicated, lack the tools for city-scale coordination.

2.2 Coordination Failure

Existing rescue channels consist primarily of WhatsApp groups, social media posts, and phone calls to NGO helplines. These methods are inefficient, prone to duplication, and offer no structured case management. A single injured animal may be reported by five different people while receiving no response due to a lack of clear assignment. Conversely, a severe case may go unnoticed because no alert mechanism exists to escalate urgency.

2.3 Absence of Data-Driven Decision Making

Local bodies and NGOs currently have no access to geo-tagged, time-stamped data on rescue incidents, abandonment patterns, or welfare outcomes. This absence prevents evidence-based resource allocation, targeted intervention, and impact measurement. PawMap addresses this by embedding spatial and temporal data collection into every user interaction, creating a living dataset that supports both operational decisions and policy advocacy.

3. Problems and Objectives

3.1 Domain and Application

PawMap operates at the intersection of Animal Welfare, Community Technology, and Geospatial Applications. The domain addresses a pressing social and public health issue that affects millions of animals and urban communities across India. The application scope encompasses the full lifecycle of stray animal intervention: initial sighting and reporting, emergency response, medical care, temporary fostering, and permanent adoption.

The scope of PawMap is intentionally broad enough to serve diverse urban environments while remaining modular, allowing city-specific NGOs or municipal bodies to deploy it with configuration adjustments. The platform is designed to be accessible to users with minimal technical literacy, making it inclusive across age groups and demographics.

The primary beneficiaries are stray and abandoned animals whose welfare outcomes improve directly through timely rescue and structured care. Secondary beneficiaries include community volunteers who gain efficient tools, NGOs that benefit from automation and reporting, veterinary professionals who access shared medical histories, potential adopters who find a trusted pipeline, and local governance bodies that receive actionable spatial intelligence.

3.2 Business Problems

The following six core problems define the business logic and feature architecture of PawMap:

No.	Problem Statement	Description
1	Untracked Stray Animal Presence	No systematic mechanism exists to log, locate, or monitor stray animals across city wards, leading to unreported cases and missed rescues.
2	Fragmented Volunteer Coordination	Multiple rescuers often respond to the same case or none at all, due to the absence of real-time assignment and conflict-prevention systems.

3	Lack of Medical Record Continuity	Animal medical history is lost between caregivers, leading to duplicate treatments, missed vaccinations, and uninformed veterinary decisions.
4	Low Adoption and Rehoming Rates	The absence of accessible, structured adoption pipelines means rescued animals remain in temporary care indefinitely with no clear path to permanent placement.
5	Absence of Behavioural Safety Data	Volunteers and the public lack standardised information about an animal's temperament, increasing the risk of injury and reducing willingness to intervene.
6	No Spatial Intelligence for Intervention	Local bodies and NGOs lack geo-analytical data on abandonment patterns and rescue outcomes, preventing evidence-based resource allocation and policy decisions.

3.3 Objectives

In response to the problems identified above, PawMap sets out the following objectives:

1. To enable any community member to report a stray or injured animal within 30 seconds by submitting a geo-tagged photograph and basic case details through the mobile or web interface.
2. To eliminate duplicate and missed responses through an atomic case-claiming system that locks a case to the first volunteer who accepts it, with automatic reassignment after 30 minutes of inactivity.
3. To maintain a continuous, caregiver-independent medical record for each animal, accessible by all verified veterinary professionals and NGO members on the platform.
4. To provide a standardised five-point temperament rating system that aggregates multi-volunteer input into a weighted safety indicator, reducing risk for first responders.
5. To facilitate structured adoption through a foster-to-adopt pipeline, adopter interest queues, and verified adopter profiles that match animals to suitable permanent homes.
6. To generate geo-analytical insights including abandonment hotspot heatmaps using DBSCAN spatial clustering and automated monthly impact reports for NGO and municipal use.

3.4 Major Technology Planned

PawMap employs a combination of cloud computing, mobile application frameworks, geospatial technology, and artificial intelligence to deliver its features. The technology choices are governed by the need for cross-platform accessibility, real-time responsiveness, and scalability:

Technology	Category	Role in PawMap
Django + Django REST Framework	Backend	Core application logic and RESTful API layer connecting the React frontend to the database and ML components

React + Bootstrap 5	Frontend	Web interface for community reporting, volunteer dashboard, and NGO/admin views
PostgreSQL + PostGIS	Cloud / Database	Relational data storage with spatial query support
Celery + Redis	Backend / Task Queue	Background job scheduling for geofenced alerts, auto-release of unclaimed cases after 30 minutes, and SOS escalation
Google Maps API	Geospatial	Live mapping, geofencing, and proximity alerts
Firebase Cloud Messaging	Cloud / Notifications	Real-time push notifications for volunteer dispatch and SOS
Python (scikit-learn)	AI / Data Science	DBSCAN clustering for hotspot mapping, coverage gap analysis, and volunteer reliability scoring

The combination of AI-driven analytics, cloud infrastructure, and geospatial services positions PawMap as a technically robust platform capable of scaling from a single-city pilot to a national deployment.

3.5 Data Science Components

While the platform as a whole spans full-stack application development, its analytical core is data science driven. As case data accumulates through community reporting, three components transform raw geo-tagged reports into actionable intelligence for volunteers, NGOs, and municipal stakeholders:

Component	Description
Hotspot Detection	DBSCAN spatial clustering applied to geo-tagged case reports (latitude, longitude, timestamp, species, case severity) to identify statistically significant abandonment clusters by ward, distinguishing genuine hotspots from areas with high reporting density due to active volunteer presence.
Coverage Gap Analysis	Spatial join between reported case locations and the geographic footprint of active, reliable volunteers to flag wards and neighbourhoods with reported cases but no responding volunteer within a defined radius, surfacing zones at risk of being structurally underserved.
Volunteer Reliability Scoring	A weighted scoring model combining response rate, case completion rate, and average response time per volunteer, used to prioritise case assignment, inform the waitlist queue, and identify volunteers eligible for SOS-tier cases.

Together, these components ensure that PawMap functions not only as a coordination tool but as a continuously improving analytical system: hotspot detection directs attention to where intervention is most needed, coverage gap analysis ensures no ward is structurally neglected, and reliability scoring keeps the volunteer network efficient and accountable as it scales.

4. Product Delivery Mode

PawMap will be delivered as a cross-platform application accessible on both mobile devices and web browsers. The mobile application, built using React Native, will support Android and iOS operating systems and will serve as the primary interface for community reporters, volunteers, and adopters who require location-aware, on-the-go access. The web application will serve NGO administrators, veterinary professionals, and municipal data analysts who require dashboard-level access and report generation on desktop devices.

No additional physical devices are required for core functionality. However, the platform is designed to be compatible with future integration of IoT-enabled tagging devices, such as NFC ear tags for sterilised or vaccinated animals, which could be incorporated in a later phase to enable passive identification and record lookup by scanning.

The backend infrastructure will be hosted on a cloud provider (AWS or Google Cloud Platform) to ensure high availability, geographic redundancy, and the ability to scale compute and storage resources in response to user growth. All personal data will be handled in compliance with applicable data protection standards.

5. References

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